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EDUCATION TO INNOVATION



Institute Name:	Silver Oak College of Computer Application		
Programme Name:	Master of Science Data Science	Semester:	1
Course Code:	DSM5061C	Course Name:	Information and Data Security
Unit No.:	2	Unit Name:	Data Privacy and Protection

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Introduction to Data Privacy and Protection



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Data privacy governs how personal information is collected and used; data protection safeguards it against loss, misuse, or unauthorized access.



Privacy

The right to control one's own personal data



Protection

Technical & organizational safeguards



Compliance

Alignment with IT Act, GDPR & DPDP Act

Personal Data, Sensitive Personal Data and Personally Identifiable Information (PII)



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Personal Data

Any data relating to an identifiable person



Sensitive Personal Data

Health, biometric, financial, religious data



PII

Data that can directly identify an individual

Examples: Name & address (Personal Data) • Medical records & biometrics (Sensitive) • Aadhaar, PAN, email (PII)

Data Privacy Principles and Privacy Requirements



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Lawfulness & Fairness

Process data with valid legal basis



Purpose Limitation

Use data only for stated purposes



Data Minimization

Collect only what is necessary



Accuracy

Keep data correct and up to date



Storage Limitation

Retain data only as long as needed



Accountability

Demonstrate compliance & governance

Data Collection and Secure Data Acquisition



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Consent

Obtain explicit user consent



Collect

Gather data via secure channels



Validate

Verify accuracy & completeness



Secure Transfer

Encrypt data in transit (TLS)

Secure acquisition ensures data is captured lawfully, minimally, and transmitted through encrypted, tamper-resistant channels.

Data Storage and Secure Data Management



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Encryption at Rest

Data encrypted on disk & databases



Access Control

Role-based restricted access



Redundancy

Replicated storage for resilience



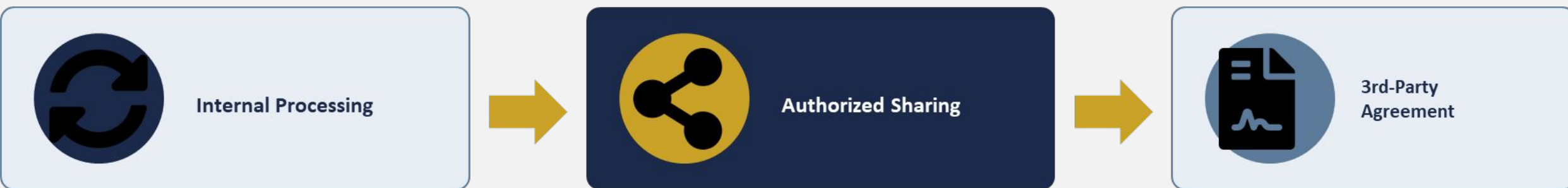
Monitoring

Continuous audit of access logs

Data Processing and Data Sharing



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- Processing must align with the original purpose of collection
- Sharing requires data processing agreements (DPAs) with third parties
- Cross-border transfers need adequacy or contractual safeguards

Data Disposal and Secure Data Destruction



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Retention Ends

Data reaches end of lifecycle



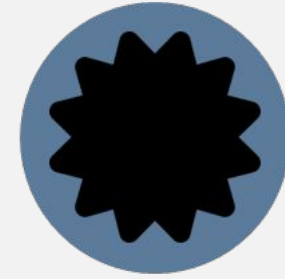
Secure Wipe

Overwrite / cryptographic erasure



Physical Destruction

Shred or degauss media



Certify

Issue certificate of destruction

Improper disposal is a leading cause of data breaches — destruction must be verifiable, irreversible, and documented.

Data Classification and Data Categorization



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Classification determines the handling, access, and protection controls applied to each data category.



Restricted

Highest sensitivity — strict controls



Confidential

Restricted to authorized roles



Internal

Limited to organization



Public

No harm if disclosed

Data Masking and Data Protection Techniques



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Original: John Smith, 4111-2233-4455-6678



J*** S****, XXXX-XXXX-XXXX-
6678



Encryption

Reversible with a key



Tokenization

Replace with non-sensitive token



Redaction

Permanently remove/black-out

Data Anonymization and Pseudonymization



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Anonymization

- Data permanently stripped of identifiers
- Cannot be re-linked to an individual
- Falls outside most privacy regulations



Pseudonymization

- Identifiers replaced with reversible tokens
- Re-identification possible with a separate key
- Still considered personal data under law

Secure File Sharing and Secure Data Storage



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Encrypted Links

Password-protected share links



Access Expiry

Time-limited download windows



Secure Cloud Storage

Encrypted, access-audited repositories



Least Privilege

Share only with authorized recipients

Data Backup, Recovery and Business Continuity



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The 3-2-1 Backup Rule

3

Copies of Data

One primary + two backups

2

Storage Media Types

e.g. disk and cloud

1

Off-site Copy

Protects against site disasters

Business Continuity Planning (BCP) ensures rapid recovery of critical data and services after disruption, minimizing downtime and data loss.

Digital Footprint and Social Media Privacy



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Active Footprint

- Posts, comments, uploaded content
- Deliberately shared information
- Directly controllable by the user



Passive Footprint

- Cookies, location, browsing history
- Collected without explicit action
- Requires privacy settings & controls

Data Privacy Challenges in Digital Analytics



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Tracking Cookies

Persistent cross-site user tracking



Behavioral Profiling

Inferred preferences from activity



Third-Party Data

Data shared with ad/analytics networks



Consent Gaps

Analytics running before consent capture

Data Protection Policies, Handling Best Practices and Data Loss Prevention (DLP) Awareness



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Written Policies

Documented data handling standards



Best Practices

Least privilege, encryption, training



DLP Monitoring

Detect & block unauthorized data
flow



Incident Response

Defined breach reporting procedure

DLP tools continuously scan endpoints, networks, and storage to prevent sensitive data from leaving the organization without authorization.



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Thank you